Integration — Lesson 29 Test: Integral Properties & Symmetry

10 questions · show your work in the box · no calculator needed · exact answers (fractions are fine)

Name: ______________________________ Date: ______________ Score: ______ / 10

1. Given $\int_0^8 f\,dx = 15$ and $\int_0^3 f\,dx = 6$, find $\int_3^8 f\,dx$. 1 pt

2. Given $\int_2^9 f\,dx = 4$, find (a) $\int_9^2 f\,dx$ and (b) $\int_5^5 f\,dx$. Name the rule you use for each part. 1 pt

3. Given $\int_1^6 f\,dx = 7$ and $\int_1^6 g\,dx = 2$, find $\int_1^6 (f + 4g)\,dx$. 1 pt

4. Classify each function as even, odd, or neither, showing the $f(-x)$ test each time: (a) $x^6 + 3$ (b) $x^5 - x$ (c) $x^3 + 1$. 1 pt

5. Evaluate $\int_{-3}^{3} x^3\,dx$ using symmetry, then verify your answer with the FTC (antiderivative and endpoints shown). 1 pt

6. Evaluate $\int_{-1}^{1} x^2\,dx$ using the even-function shortcut, then verify by running the FTC straight across from −1 to 1. 1 pt

7. Evaluate $\int_{-2}^{2} (x^5 + 3)\,dx$ by splitting into terms and using the right rule on each term. 1 pt

8. A function g is EVEN, and $\int_{0}^{5} g\,dx = 9$. Find (a) $\int_{-5}^{5} g\,dx$ and (b) $\int_{-5}^{0} g\,dx$. 1 pt

9. A function f is ODD, and $\int_0^4 f\,dx = 3$. Find (a) $\int_{-4}^4 f\,dx$, (b) $\int_{-4}^0 f\,dx$, and (c) the average value of f on $[-4, 4]$. 1 pt

10. A student writes: "$\int_{-2}^2 x^3\,dx = 2 \cdot \int_0^2 x^3\,dx = 2 \cdot 4 = 8$." Explain exactly what went wrong, give the correct value, and verify it with the FTC. 1 pt